United States Patent Application Publication	20250277831
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	PENG; Wei-Hsuan

ENERGY MEASUREMENT DEVICE

Abstract

An energy measurement device comprising a conversion circuit and a first measurement circuit. The conversion circuit is coupled to a sensor through an input port of the energy measurement device to receive a sensing voltage. The conversion circuit is configured to convert the sensing voltage into a preset measurement signal, and provide the preset measurement signal to the processing circuit. The first measurement circuit is coupled to a processing circuit through the conversion circuit, and is coupled to the sensor through the input port. The first measurement circuit comprises a first switch, when the first switch receives a first enable signal provided by the processing circuit, the first measurement circuit is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

Inventors: **PENG; Wei-Hsuan (New Taipei City, TW)**

Applicant: **Chicony Power Technology Co., Ltd.** (New Taipei City, TW)

Family ID: **96853191**

Appl. No.: **18/665598**

Filed: **May 16, 2024**

Foreign Application Priority Data

TW	113107598	Mar. 01, 2024

Publication Classification

Int. Cl.: **G01R22/10** (20060101); **G01R19/25** (20060101)

U.S. Cl.:

CPC **G01R22/10** (20130101); **G01R19/2506** (20130101); **G01R19/2513** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Taiwan Application Serial Number 113107598, filed Mar. 1, 2024, which is herein incorporated by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to energy measurement technology, and in particular to an energy measurement device.

Description of Related Art

[0003] With the increasingly severe issue of global warming and the accelerated depletion of energy resources, “energy conservation and carbon reduction” has become an issue that has received more and more attention in recent years. From the government to businesses, efforts have been devoted to developing green energy to create a sustainable low-carbon society and economy.

[0004] To achieve energy conservation and carbon reduction, in order to know how to improve energy consumption problems, it is essential to have a clear understanding of the historical records of energy consumption in order to identify how to improve energy efficiency issues. Therefore, designing an energy measurement device that can be applied to different energy-consuming equipment and environments has become a major current challenge.

SUMMARY

[0005] One aspect of the present disclosure is an energy measurement device, comprising a conversion circuit and a first measurement circuit. The conversion circuit is coupled to a processing circuit, and is coupled to a sensor through an input port of the energy measurement device to receive a sensing voltage. The conversion circuit is configured to convert the sensing voltage into a preset measurement signal, and provide the preset measurement signal to the processing circuit. The first measurement circuit is coupled to the processing circuit through the conversion circuit, and is coupled to the sensor through the input port. The first measurement circuit comprises a first switch, when the first switch receives a first enable signal provided by the processing circuit, the first measurement circuit is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

[0006] Another aspect of the present disclosure is an energy measurement device, comprising a processing circuit, a conversion circuit and a first measurement circuit. The conversion circuit is coupled to the processing circuit, and is coupled to a sensor through an input port of the energy measurement device to receive a sensing voltage. The first measurement circuit is coupled to the processing circuit through the conversion circuit, and is coupled to the sensor through the input port. When the processing circuit disables the first measurement circuit, the conversion circuit is configured to convert the sensing voltage into a preset measurement signal, and provide the preset measurement signal to the processing circuit. When the processing circuit enables the first measurement circuit, the first measurement circuit is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

[0007] Another aspect of the present disclosure is an energy measurement device, comprising a processor, a plurality of input ports and a plurality of energy measurement circuits. The plurality of input ports is coupled to a plurality of sensors. Each of the plurality of energy measurement circuits is respectively coupled to one of the plurality of input ports, and comprises a conversion circuit and a first measurement. The conversion circuit is coupled to the processor, and is coupled to a corresponding one of the plurality of input ports to receive a sensing voltage of a corresponding

one of the plurality of sensors. When the sensing voltage is within a preset voltage range, the conversion circuit is configured to convert the sensing voltage into a preset measurement signal. The first measurement circuit is coupled to the processor through the conversion circuit, and is coupled to the corresponding one of the plurality of input ports. When the sensing voltage is within a first voltage range, the first measurement circuit is enabled by the processor, and is configured to convert the sensing voltage into a first measurement voltage. The conversion circuit is configured to convert the first measurement voltage into a first measurement signal.
[0008] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
[0010] FIG. **1** is a schematic diagram of an energy management system in some embodiments of the present disclosure.
[0011] FIG. **2** is a schematic diagram of an energy measurement device in some embodiments of the present disclosure.
[0012] FIG. **3** is a functional block diagram of the energy measurement device in some embodiments of the present disclosure.
[0013] FIGS. **4**A-**4**C are schematic diagrams of the operations of the energy measurement device in some embodiments of the present disclosure.
DETAILED DESCRIPTION
[0014] For the embodiment below is described in detail with the accompanying drawings, embodiments are not provided to limit the scope of the present disclosure. Moreover, the operation of the described structure is not for limiting the order of implementation. Any device with equivalent functions that is produced from a structure formed by a recombination of elements is all covered by the scope of the present disclosure. Drawings are for the purpose of illustration only, and not plotted in accordance with the original size.
[0015] It will be understood that when an element is referred to as being “connected to” or “coupled to”, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element to another element is referred to as being “directly connected” or “directly coupled,” there are no intervening elements present. As used herein, the term “and/or” includes an associated listed items or any and all combinations of more.
[0016] The represent disclosure relates to an energy measurement device, which can be applied to various load devices. The energy measurement device is coupled to a sensor arranged on a load device to receive and record an energy consumption data of the load device. Depending on the type of the load device, the information categories that present the energy consumption data may also be different. For example, when detecting one load device, the sensor records “voltage” of the load device to present the current energy consumption status. When detecting another load device, the sensor records “current” of the load device to present the current energy consumption status. In other words, sensor can output different types of energy consumption data.
[0017] In one embodiment, different types of energy consumption data must be recorded by specific energy measurement devices, or received through specific receiving ports. For example, for a sensor that outputs “voltage”, the energy measurement device needs to receive it through a specific voltage receiving port, and for a sensor that outputs “current”, the energy measurement device needs to receive it through a specific current receiving port. Therefore, the arrangement of

conventional energy measurement devices is relatively inconvenient and has high arrangement costs.

[0018] In other embodiments, conventional energy measurement device is equipped with a switch, and user can control the switch according to different types of the energy consumption data to select different receiving circuits in the energy measurement device to receive signals. However, this method still requires manual operation by the user, so it is not perfect in use.

[0019] FIG. **1** is a schematic diagram of an energy management system in some embodiments of the present disclosure. The energy management system **100** includes multiple sensors **110** and multiple energy measurement devices **120**. Each of sensors **110** is respectively arranged on a load device **130** to obtain/capture electrical data (e.g., voltage, current) of the load device **130** and generates a sensing signal.

[0020] The energy measurement device **120** is coupled to one or multiple sensors **110** to receive a sensing signal provided by the sensors **110**. Due to sensing signals being transmitted to the energy measurement device **120** in the form of electrical signal, and since voltage and current are two sides of the same electrical signal, the sensing signal(s) provided by the sensors **110** is referred to “sensing voltage” herein.

[0021] In one embodiment, the energy management system **100** further includes a control device **140**, a management server **150** and a network equipment **160**. The control device **140** can be a portable device (e.g., smart phone) that is communicatively connected to the energy measurement device **120** to manage the energy measurement device **120** (e.g., check data, set detection parameters). The management server **150** establishes a communication connection with the energy measurement device **120** in a wired or wireless manner to receive all data obtained by the energy measurement device **120** and integrate it into energy consumption records for users to check at any time. As shown in the figure, the energy measurement device **120** may be indirectly connected to the management server **150** through the network equipment **160**, or directly connected to the management server **150**. Since people in the art can understand the method of transmitting information between devices through communication technology, and thus they are not further detailed herein.

[0022] FIG. **2** is a schematic diagram of an energy measurement device in some embodiments of the present disclosure, which can be applied to implement one of the energy measurement devices **120** shown in FIG. **1**. In one embodiment, the energy measurement device **200** at least includes a input port **210**, a conversion circuit **220**, a first measurement circuit **230** and a processing circuit **240**.

[0023] Referring to FIG. **1** and FIG. **2**, the input port **210** is coupled to the sensor **110** to receive a sensing voltage Vin from the sensor **110**. In order to make the diagram concise, the input port **210** is represented by “node” here. It should be mentioned here that although only one input port **210** is shown in FIG. **2**, in other embodiments, the energy measurement device **200** may include multiple input ports **210**, and each of input ports **210** is configured to be coupled to a different sensor **110**. In addition, each input port **210** corresponds to a circuit structure shown in FIG. **2** to receive the sensing voltage Vin provided by the corresponding sensor **110** respectively. The connection specifications of the input port **210** can be changed according to the actual specifications of the sensor **110**. Since people in the art can understand the implementation of input ports **210**, and thus they are not further detailed herein.

[0024] The input terminal of the conversion circuit **220** is coupled to the sensor **110** through the corresponding input port **210** to receive the sensing voltage Vin provided by the sensor **110**. The output terminal of the conversion circuit **220** is coupled to the processing circuit **240**. As shown in FIG. **2**, the conversion circuit **220** is coupled between the input port **210** and the processing circuit **240**, and is configured to convert the sensing voltage Vin into a preset measurement signal, and provide the preset measurement signal to the processing circuit **240**.

[0025] In one embodiment, the conversion circuit **220** includes a voltage divider circuit and a filter circuit, and is configured to divide and filter the sensing voltage Vin, so that the converted preset measurement signal is within a preset measurement range. In some embodiments, when the sensing voltage Vin is within the preset voltage range, the energy measurement device **200** chooses to use the conversion circuit **220** to convert the sensing voltage Vin. For example, the conversion circuit **220** is configured to receive and convert the sensing voltage Vin of an industrial motor, when the industrial motor operating, the sensing voltage Vin detected by the sensor **110** will be between “10-20 volts” (i.e., a preset voltage range), and the preset measurement signal generated by the conversion circuit **220** will be between “3-6 volts” (i.e., a preset measurement range). Multiple methods for the energy measurement device **200** to receive different signals by selecting different circuits will be explained in subsequent paragraphs.

[0026] Specifically, the conversion circuit **220** includes a voltage limiting element D**21**, multiple voltage dividing resistors R**21**-R**22** and a filter capacitor C**21** to implement the voltage divider circuit and the filter circuit. The voltage limiting element D**21** is coupled to the input port **210**, and is configured to stabilize the voltage of the input terminal of the conversion circuit **220**. The voltage limiting element D**21** can be realized by a Zener diode.

[0027] The voltage dividing resistors R**21**-R**22** are coupled to the voltage limiting element D**21**, and are configured to divides the sensing voltage Vin. In one embodiment, the voltage dividing resistor R**21** is coupled between the input port **210** and the processing circuit **240**. The voltage dividing resistor R**22** is coupled between the voltage dividing resistors R**21** and a reference potential (e.g., groung). The filter capacitor C**21** is coupled between the voltage dividing resistors R**21**-R**22** and the processing circuit **240**, and is configured to filter out noise in the voltage.

[0028] The first measurement circuit **230** is coupled to the sensor **110** through the input port **210**, and is coupled to the processing circuit **240** through the conversion circuit **220**. The processing circuit **240** is configured to selectively enable or disable the first measurement circuit **230**, so that the sensing voltage Vin is converted into different measurement signals by different circuits.

[0029] In one embodiment, the first measurement circuit **230** includes a first switch T**21**. The turned-on or turned-off state of the first switch T**21** is controlled by a control signal Sc**1** provided by the processing circuit **240**. For example, when the processing circuit **240** transmits a first enable signal to a control terminal of the first switch T**21**, the first switch T**21** will be in a turned-on state. When the processing circuit **240** transmits a first disable signal to the control terminal of the first switch T**21**, the first switch T**21** will be in the turned-off state, and the first measurement circuit **230** will become open. In one embodiment, the first switch T**21** can be implemented by an N-type or P-type Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET), but in some other embodiments, it can also be implemented by other transistor switches or switching elements.

[0030] As shown in FIG. **2**, when the processing circuit **240** disables the first measurement circuit **230**, since the first measurement circuit **230** becomes open (switched to open circuit), the first measurement circuit **230** will not perform any processing on the sensing voltage Vin. At this time, the conversion circuit **220** will directly receive the sensing voltage Vin and convert the sensing voltage Vin into the preset measurement signal.

[0031] On the other hand, when the processing circuit **240** enables the first measurement circuit **230**, the first measurement circuit **230** can operate normally. Since the first measurement circuit **230** is coupled between the input port **210** and the conversion circuit **220**, the sensing voltage Vin first will be converted into the first measurement voltage by the first measurement circuit **230**. Then, the conversion circuit **220** converts the first measurement voltage (instead of the original sensing voltage Vin) to generate a first measurement signal.

[0032] Accordingly, by selectively enabling or disabling the first measurement circuit **230**, the energy measurement device **200** can convert the sensing voltage Vin by different circuits or circuit combinations according to different sensing conditions. For example, in the case that the load device **130** coupled to the input port **210** and the sensor **110** is an industrial motor, the energy measurement device **200** converts the sensing voltage Vin by the conversion circuit **220** to correctly interpret the current energy consumption of the industrial motor. In the case that the load device **130** coupled to the input port **210** and the sensor **110** is an industrial water pump, the energy measurement device **200** converts the sensing voltage Vin by the by the conversion circuit **220** and the first measurement circuit **230** to correctly interpret the current energy consumption of the industrial water pump.

[0033] To facilitate understanding the application and effectiveness of the energy measurement device of the represent disclosure, an explanation is provided herein using FIG. **3** as an example. FIG. **3** is a functional block diagram of an energy measurement device **300** in some embodiments of the present disclosure, which can be implemented the energy measurement device **120** shown in FIG. **1**, and the energy measurement device **200** shown in FIG. **2** can be implemented as a part of the energy measurement device **300**.

[0034] As shown in FIG. **3**, the energy measurement device **300** includes a main meter **310** and one or multiple extended meters **320**. The main meter **310** includes a controller **311**, and is coupled to the extended meter(s) **320**. The extended meter **320** includes a processor **321**, multiple energy measurement circuits **322** and multiple input ports **323**. Each of the input ports **323** respectively corresponds and is coupled to a different energy measurement circuit **322**, and each of the energy measurement circuits **322** respectively corresponds to a different sensor **330**. The energy measurement circuit **322** can be implemented by the conversion circuit **220**, the first measurement circuit **230** and the second measurement circuit **250** shown in FIG. **2**.

[0035] In one embodiment, the processor **321** can be the processing circuit **240** shown in FIG. **2**. In other words, the processor **321** (the processing circuit **240**) is configured to receive measurement voltages V**31**-V**34** transmitted by different energy measurement circuits **322**. In some other embodiments, the operation performed by the processing circuit **240** shown in FIG. **2** can be implemented by the controller **311** and the processor **321** shown in FIG. **3**.

[0036] In one embodiment, the extended meter **320** is coupled to multiple sensors **330** through multiple input ports **323**, and each sensor **330** is further coupled to the different load device (not shown in FIG. **3**). The extended meter **320** interpret the received measurement voltages V**31**-V**34** through the energy measurement circuit **322** as multiple measurement signals S**31**-S**34**, and provides the measurement signals S**31**-S**34** to the main meter **310**. In addition, the main meter **310** is configured to provide driving power DP to the extended meter **320**, so that the extended meter **320** obtains the driving power DP for operation.

[0037] In one embodiment, multiple extended meters **320** are connected in series with each other. Therefore, the driving power DP provided by the main meter **310** will first pass through the first extended meter **320** and then be provided to the second extended meter **320**. Similarly, the measurement signal S**35** obtained by the second extended meter **320** will first pass through the first extended meter **320** and then be transmitted to the main meter **310**.

[0038] As mentioned above, the energy measurement device **120**/**200**/**300** includes the conversion circuit and one or more measurement circuits, and can receive and convert the sensing voltage Vin by different circuit combinations to correctly interpret the measurement signal. In other words, for different load devices, the conversion circuit and the measurement signal generated by each measurement circuit will correspond to different measurement ranges.

[0039] FIG. **4**A-**4**C are schematic diagrams of the operations of the energy measurement device in some embodiments of the present disclosure. Here, the circuit architecture of the energy measurement device **200** is used to describe the circuit details and operation. As mentioned before, since voltage and current are two sides of an electrical signal, the circuit itself cannot directly determine whether the voltage or current should be recorded. The energy measurement device **200** can automatically select different circuits or circuit combinations to convert/interpret the sensing voltage Vin by being set in different operating modes (e.g., remote configuration via the control device **140** shown in FIG. **1**).

[0040] As shown in FIG. **4**A, in one embodiment, the conversion circuit **220** is configured to detect the sensing signal which is a voltage type (i.e., using a voltage value to represent a sensing data). In other words, when the energy measurement device **200** is set to a voltage mode, the energy measurement device **200** disable other circuits, and convert the sensing voltage Vin only by the conversion circuit **220** to generate the preset measurement signal S**41**.

[0041] In one embodiment, the voltage mode is a preset mode. In another embodiment, the processing circuit **240** disable other circuits (e.g., the first measurement circuit **230**, the second measurement circuit **250**) according to a voltage mode signal. In other words, when the conversion circuit **220** directly convert the sensing voltage Vin into the preset measurement signal S**41**, the first measurement circuit **230** and the second measurement circuit **250** are disabled and in open circuit state, so the sensing voltage Vin will not be affected.

[0042] The first measurement circuit **230** is configured to detect the sensing signal which is a current type (i.e., using a current value to represent a sensing data). In other words, when the energy measurement device **200** is set to a current mode, the energy measurement device **200** enables the first measurement circuit **230**, and disables other circuits (e.g., disables the second measurement circuit **250**).

[0043] As shown in FIG. **4**B, in one embodiment, the first measurement circuit **230** includes one or more multiple first impedance elements R**23**-R**24**. When the first switch T**21** receives the first enable signal provided by the processing circuit **240**, the sensing voltage Vin forms cross-voltages on the first impedance elements R**23**-R**24** (i.e., convert current characteristics into voltage characteristics to present). The first measurement circuit **230** generates a first measurement voltage S**42** according to cross-voltages of the first impedance elements R**23**-R**24**. At this time, since a voltage of the input terminal of the conversion circuit **220** will be changed to the first measurement voltage S**42** by the first measurement circuit **230**, the conversion circuit **220** converts the first measurement voltage S**42** to output a first measurement signal S**43** to the processing circuit **240**.

[0044] In the embodiment shown in FIG. **4**B, the first measurement circuit **230** includes multiple first impedance elements R**23**-R**24**, the first impedance elements R**23**-R**24** are connected in parallel to form multiple shunt currents. Accordingly, the sensing voltage Vin will be prevented from causing damage to the first impedance elements R**23**-R**24** when the sensing voltage Vin has an abnormally high voltage. In addition, electrical stress of the first measurement circuit **230** can be improved. In some other embodiments, the first measurement circuit **230** can only have one impedance element to form the first measurement voltage S**42**.

[0045] As shown in FIGS. **1**, **2** and **4**C, in one embodiment, the energy measurement device **200** further includes a second measurement circuit **250**. The second measurement circuit **250** is coupled to the processing circuit **240** through the conversion circuit **220**, and is coupled to the sensor **110** through the input port **210**. The second measurement circuit **250** includes a second switch T**22**. The second switch T**22** is coupled between a bias voltage Vd**1** and the input terminal of the conversion circuit **220**. The turned-on or turned-off state of the first switch T**22** is controlled by a control signal Sc**2** provided by the processing circuit **240**. For example, when the processing circuit **240** transmits a second enable signal to a control terminal of the second switch T**22**, the second switch T**22** will be in a turned-on state. When the processing circuit **240** transmits a second disable signal to the control terminal of the second switch T**22**, the second switch T**22** will be in the turned-off state, and the second measurement circuit **250** will become open (switched to open circuit). In one embodiment, the second switch T**22** can be implemented by an N-type or P-type Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET), but in some other embodiments, it can also be implemented by other transistor switches or switching elements.

[0046] The second measurement circuit **250** is configured to detect the sensing signal which is a temperature type (i.e., using temperature to represent a sensing data). “Temperature type” means that the load device/sensor has a thermistor, which can convert the temperature into an impedance value. As shown in FIG. **4**C, when detecting the sensing signal of the temperature type, an impedance value of a thermistor R**41** of the sensor **110** changes with the temperature of the sensor

110 or the load device. In one embodiment, a temperature coefficient of the thermistor R**41** is negative (ie, Negative Temperature Coefficient, hereinafter referred to as NTC). In other words, the higher the temperature, the smaller the impedance value.

[0047] When the energy measurement device **200** is set to a temperature mode (NTC mode), the energy measurement device **200** enables the second measurement circuit **250**, and disables other circuits (e.g., the first measurement circuit **230**).

[0048] As mentioned above, when the second switch T**22** receives a second enable signal provided by the processing circuit **240**, the second measurement circuit **250** is configured to convert the sensing voltage Vin into a second measurement voltage S**44**. At the same time, the first measurement circuit **230** receives the first disable signal provided by the processing circuit **240**, and then the first measurement circuit **230** is disabled and becomes open (switched to open circuit). Since a voltage of the input terminal of the conversion circuit **220** will be changed by the second measurement circuit **250** to the second measurement voltage S**44**, so the conversion circuit **220** converts the second measurement voltage S**44** to output a second measurement signal S**45** to the processing circuit **240**.

[0049] As shown in FIG. **4**C, in one embodiment, the second measurement circuit **250** further includes a second impedance element R**25**. When the second switch T**22** receives the second enable signal to enable the second measurement circuit **250**, the second measurement circuit **250** uses the second impedance element R**25** and the thermistor R**41** of the sensor **110** to divide the sensing voltage Vin. A voltage division result will generate the second measurement voltage S**44** on the input terminal of the conversion circuit **220**. Accordingly, through a voltage dividing characteristics of resistors, “impedance” characteristics will be converted into “voltage” characteristics.

[0050] As mentioned above, the energy measurement device **200** can convert the sensing voltage Vin by different circuits or circuit combinations according to different conditions. For example, when the sensing voltage is within the preset voltage range, the energy measurement device **200** converts the sensing voltage Vin by the conversion circuit **220**. When the sensing voltage is within the first voltage range, the energy measurement device **200** converts the sensing voltage Vin by the conversion circuit **220** and the first measurement circuit **230**. When the sensing voltage is within the second voltage range, the energy measurement device **200** converts the sensing voltage Vin by the conversion circuit **220** and the second measurement circuit **250**. Each voltage range is different from each other.

[0051] In addition, in one embodiment, the energy measurement device further includes a level shifter **260**. The level shifter **260** is coupled to a bias voltage Vd**2**, the processing circuit **240** and the second switch T**22**, and is configured to receive a control signal Sc**2** (e.g., the second enable signal or the second disable signal) provided by the processing circuit **240**. The level shifter **260** is coupled to convert the control signal Sc**2** to a suitable voltage level to control the turned-on or turned-off of the second switch T**22**. Since people in the art can understand the operating principles and changes of the level shifter, and thus they are not further detailed herein.

[0052] In the aforementioned embodiments, the energy measurement device **120**/**200**/**300** can be set in different modes to selectively convert the sensing voltage Vin by different circuits or circuit combinations. In some other embodiments, the energy measurement device **120**/**200**/**300** can have an automatic determination mechanism. To facilitate understanding, here taking three different load devices as examples and illustrate them with FIGS. **1**, **2** and **4**A-**4**C. For example, the load device of the energy management system **100** has the following three types: [0053] (1) Industrial motor: the electrical signal used by the industrial motor to record the energy consumption status is “voltage”. In an normal operation, the sensing voltage provided by the sensor is between “10-20 volts” (i.e., the preset voltage range). For the industrial motor, the energy measurement device **200** should operate in “voltage mode”. That is, other measurement circuits are disabled, and the sensing voltage after conversion will be between “3-6 volts” (i.e., the preset measurement range). [0054] (2) Distribution equipment: the electrical signal used by the distribution equipment to record the

energy consumption status is “current”. In an normal operation, the sensing current provided by the sensor forms a sensing voltage on the input port **210**, and is between “2-5 volts” (i.e., the first voltage range). For the distribution equipment, the energy measurement device **200** should operate in “current mode”. That is, the first measurement circuit is enabled, and other measurement circuits are disabled (e.g., disables the second measurement circuit **250**), and the sensing voltage after conversion will be between “1-5 volts” (i.e., a first measurement range). [0055] (3) Temperature regulator: the electrical signal used by the temperature regulator to record the energy consumption status is “an impedance value generated by the thermistor”. In an normal operation, the sensing voltage provided by the sensor is 0 volt or less than 0.1 volts (i.e., the second voltage range). For the temperature regulator, the energy measurement device **200** should operate in “NTC mode”. That is, the second measurement circuit is enabled, and other measurement circuits are disabled, and the sensing voltage after conversion will be between “1-10 volts” (i.e., a second measurement range).
[0056] As shown in FIG. **2** and FIGS. **4**A-**4**C, the energy measurement device **200** operates in the voltage mode by default. In other words, when the energy measurement device **200** receives the sensing voltage Vin, the processing circuit **240** first disables other measurement circuits (e.g., the first measurement circuit **230**, the second measurement circuit **250**). Then, after the conversion circuit **220** converts the sensing voltage Vin into the preset measurement signal S**41**, the processing circuit **240** determines whether the preset measurement signal S**41** exceeds the preset measurement range.
[0057] If the preset measurement signal S**41** exceeds the preset measurement range (e.g., 8 volts), it means that the sensing voltage Vin received at this time does not correspond to “the industrial motor”. At this time, the energy measurement device **200** will be changed to other modes (e.g., the current mode or the NTC mode) to convert the sensing voltage Vin by the first measurement circuit **230** or the second measurement circuit **250**. For example, the energy measurement device **200** is changed to the current mode, the processing circuit **240** is configured to enable the first measurement circuit **230**, but disables other measurement circuits (e.g., the second measurement circuit **250**). Accordingly, by determining whether the signal received by the processing circuit **240** is within each measurement range, the energy measurement device **200** can be able to automatically change to different modes to accurately record the energy consumption data.
[0058] The elements, method steps, or technical features in the foregoing embodiments may be combined with each other, and are not limited to the order of the specification description or the order of the drawings in the present disclosure.
[0059] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this present disclosure provided they fall within the scope of the following claims.

Claims

1. An energy measurement device, comprising: a conversion circuit coupled to a processing circuit, and coupled to a sensor through an input port of the energy measurement device to receive a sensing voltage, wherein the conversion circuit is configured to convert the sensing voltage into a preset measurement signal, and provide the preset measurement signal to the processing circuit; and a first measurement circuit coupled to the processing circuit through the conversion circuit, and coupled to the sensor through the input port, wherein the first measurement circuit comprises a first switch, when the first switch receives a first enable signal provided by the processing circuit, the first measurement circuit is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

2. The energy measurement device of claim 1, wherein the first measurement circuit further comprises a first impedance element, when the first switch receives the first enable signal, the first measurement circuit generates the first measurement voltage according to a cross-voltage of the first impedance element; when the first switch receives a first disable signal provided by the processing circuit, the first measurement circuit becomes open.

3. The energy measurement device of claim 1, further comprising: a second measurement circuit coupled to the processing circuit through the conversion circuit, and coupled to the sensor through the input port, wherein the second measurement circuit comprises a second switch, when the second switch receives a second enable signal provided by the processing circuit, the second measurement circuit is configured to convert the sensing voltage into a second measurement voltage, and the conversion circuit is configured to convert the second measurement voltage into a second measurement signal; and wherein when the second switch receives a second disable signal provided by the processing circuit, the second measurement circuit is switched to open.

4. The energy measurement device of claim 3, wherein when the conversion circuit converts the sensing voltage into the preset measurement signal, the first measurement circuit and the second measurement circuit are disabled.

5. The energy measurement device of claim 4, wherein the second measurement circuit further comprises a second impedance element, when the second switch receives the second enable signal, the second measurement circuit divides the sensing voltage according to the second impedance element and a thermistor of the sensor to generate the second measurement voltage.

6. The energy measurement device of claim 1, wherein the conversion circuit comprises: a voltage limiting element coupled to the input port; a plurality of voltage dividing resistors coupled to the voltage limiting element; and a filter capacitor coupled between the plurality of voltage dividing resistors and the processing circuit.

7. An energy measurement device, comprising; a processing circuit; a conversion circuit coupled to the processing circuit, and coupled to a sensor through an input port of the energy measurement device to receive a sensing voltage; and a first measurement circuit coupled to the processing circuit through the conversion circuit, and coupled to the sensor through the input port; wherein when the processing circuit disables the first measurement circuit, the conversion circuit is configured to convert the sensing voltage into a preset measurement signal, and provide the preset measurement signal to the processing circuit; wherein when the processing circuit enables the first measurement circuit, the first measurement circuit is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

8. The energy measurement device of claim 7, wherein when the energy measurement device receives the sensing voltage, the processing circuit first disables the first measurement circuit, and determines whether the preset measurement signal provided by the conversion circuit exceeds a preset measurement range; wherein when the preset measurement signal exceeds the preset measurement range, the processing circuit is configured to enable the first measurement circuit.

9. The energy measurement device of claim 7, wherein the first measurement circuit further comprises a first impedance element and a first switch, when the first switch receives a first enable signal provided by the processing circuit, the first measurement circuit generates the first measurement voltage according to a cross-voltage of the first impedance element; when the first switch receives a first disable signal provided by the processing circuit, the first measurement circuit becomes open.

10. The energy measurement device of claim 7, further comprising: a second measurement circuit coupled to the processing circuit through the conversion circuit, and coupled to the sensor through the input port, wherein the second measurement circuit comprises a second switch, when the second switch receives a second enable signal provided by the processing circuit, the second measurement circuit is configured to convert the sensing voltage into a second measurement voltage, and the conversion circuit is configured to convert the second measurement voltage into a second measurement signal; and wherein when the second switch receives a second disable signal provided by the processing circuit, the second measurement circuit is switched to open.

11. The energy measurement device of claim 10, wherein the processing circuit is configured to disable the first measurement circuit and the second measurement circuit according to a voltage mode signal, so that the conversion circuit converts the sensing voltage into the preset measurement signal.

12. The energy measurement device of claim 11, wherein the second measurement circuit further comprises a second impedance element, when the second switch receives the second enable signal, the second measurement circuit divides the sensing voltage according to the second impedance element and a thermistor of the sensor to generate the second measurement voltage.

13. The energy measurement device of claim 7, wherein the conversion circuit comprises: a voltage limiting element coupled to the input port; a plurality of voltage dividing resistors coupled to the voltage limiting element; and a filter capacitor coupled between the plurality of voltage dividing resistors and the processing circuit.

14. An energy measurement device, comprising; a processor; a plurality of input ports coupled to a plurality of sensors; and a plurality of energy measurement circuits, wherein each of the plurality of energy measurement circuits is respectively coupled to one of the plurality of input ports, and comprises: a conversion circuit coupled to the processor, and coupled to a corresponding one of the plurality of input ports to receive a sensing voltage of a corresponding one of the plurality of sensors, wherein when the sensing voltage is within a preset voltage range, the conversion circuit is configured to convert the sensing voltage into a preset measurement signal; and a first measurement circuit coupled to the processor through the conversion circuit, and coupled to the corresponding one of the plurality of input ports, wherein when the sensing voltage is within a first voltage range, the first measurement circuit is enabled by the processor, is configured to convert the sensing voltage into a first measurement voltage, and the conversion circuit is configured to convert the first measurement voltage into a first measurement signal.

15. The energy measurement device of claim 14, wherein when one of the plurality of energy measurement circuits receives the sensing voltage, the processor first disables the first measurement circuit, and determines whether the preset measurement signal provided by the conversion circuit exceeds a preset measurement range; wherein when the preset measurement signal exceeds the preset measurement range, the processor is configured to enable the first measurement circuit.

16. The energy measurement device of claim 14, wherein the first measurement circuit further comprises a first impedance element and a first switch, when the first switch receives a first enable signal provided by the processor, the first measurement circuit generates the first measurement voltage according to a cross-voltage of the first impedance element; when the first switch receives a first disable signal provided by the processor, the first measurement circuit becomes open.

17. The energy measurement device of claim 14, further comprising: a second measurement circuit coupled to the processor through the conversion circuit, and coupled to the corresponding one of the plurality of input ports, wherein the second measurement circuit comprises a second switch, when the sensing voltage is within a second voltage range, the second switch receives a second enable signal provided by the processor, the second measurement circuit is configured to convert the sensing voltage into a second measurement voltage, and the conversion circuit is configured to convert the second measurement voltage into a second measurement signal; and wherein the second voltage range is different from the first voltage range.

18. The energy measurement device of claim 17, wherein the processor is configured to disable the first measurement circuit and the second measurement circuit according to a voltage mode signal, so that the conversion circuit converts the sensing voltage into the preset measurement signal; and wherein when the processor provides the second enable signal to the second switch, the processor disables the first measurement circuit.

19. The energy measurement device of claim 18, wherein the second measurement circuit further comprises a second impedance element, when the second switch receives the second enable signal, the second measurement circuit divides the sensing voltage according to the second impedance element and a thermistor to generate the second measurement voltage; and wherein when the second switch receives a second disable signal provided by the processor, the second measurement circuit becomes open.

20. The energy measurement device of claim 14, wherein the conversion circuit comprises: a voltage limiting element; a plurality of voltage dividing resistors coupled to the voltage limiting element; and a filter capacitor coupled between the plurality of voltage dividing resistors and the processor.